

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25_ajust.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-08 09:34:40

Training Start: 2025-05-08_09-16-33

Training End: 2025-05-08_09-30-33

Training Duration: 14.00 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: relu

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.9775

MSE: 6.2756

RMSE: 2.5051

R2 Score: -0.0742

Multi-Step Forecast:

Forecast Steps: 20

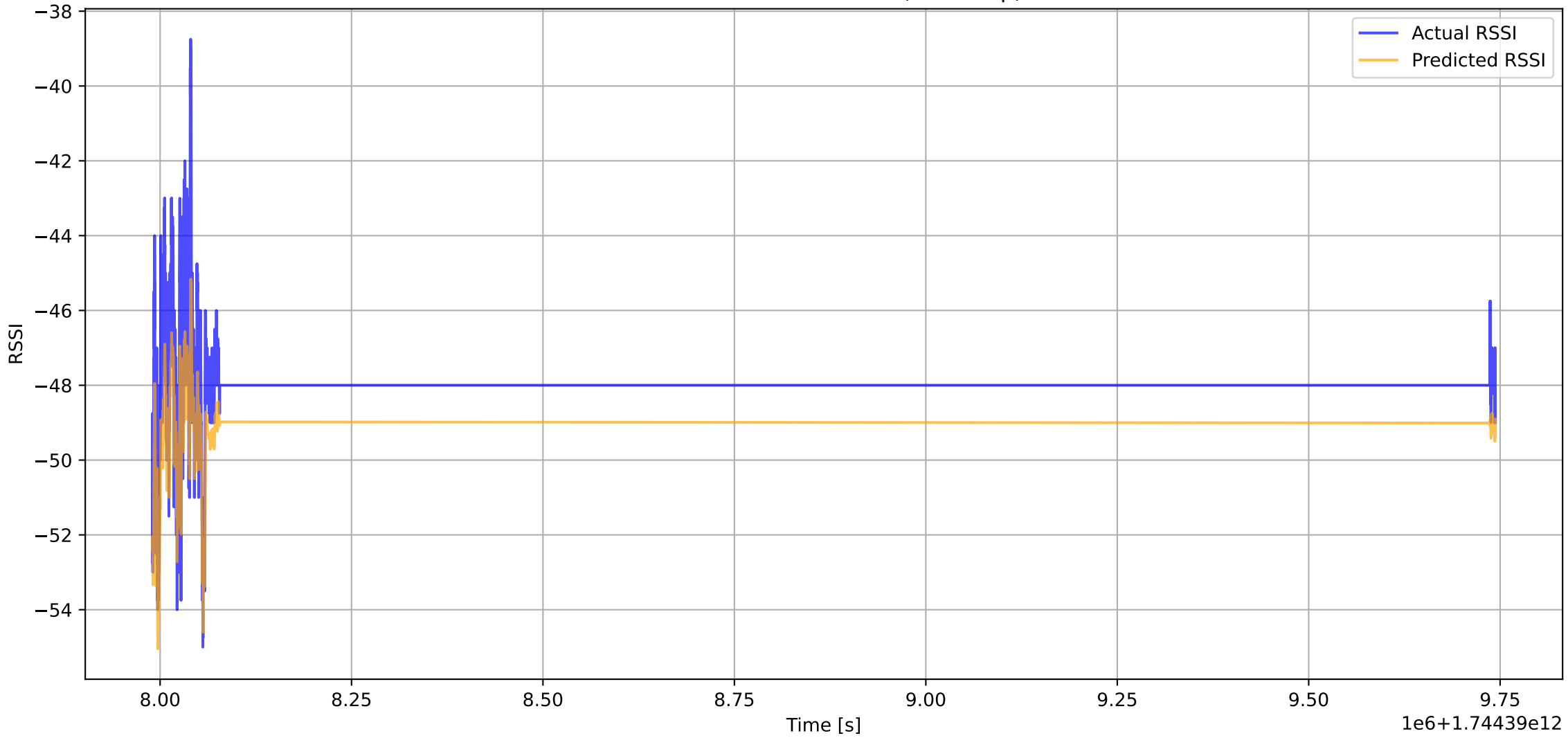
MAE: 0.0629

MSE: 0.0057

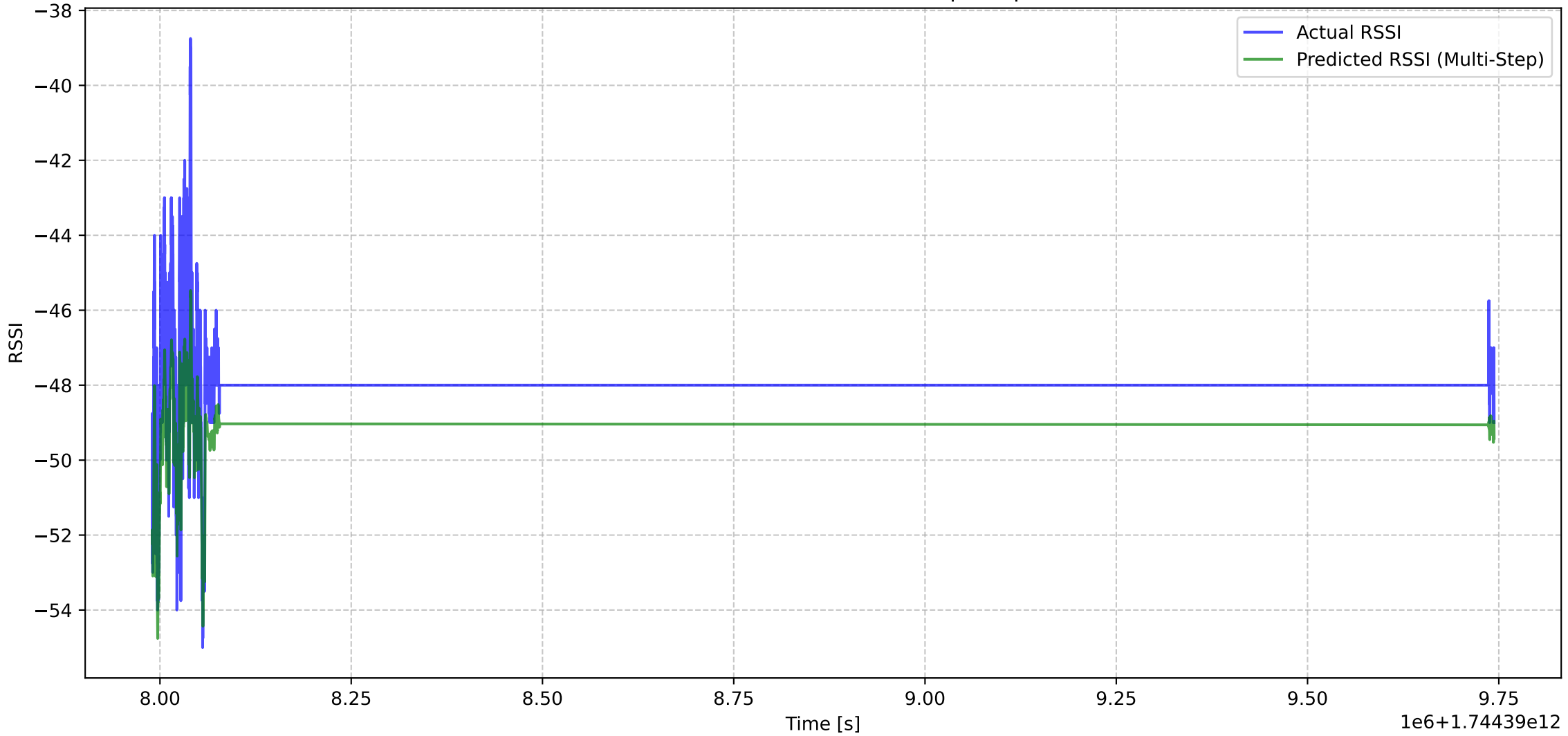
RMSE: 0.0755

R2 Score: 0.2346

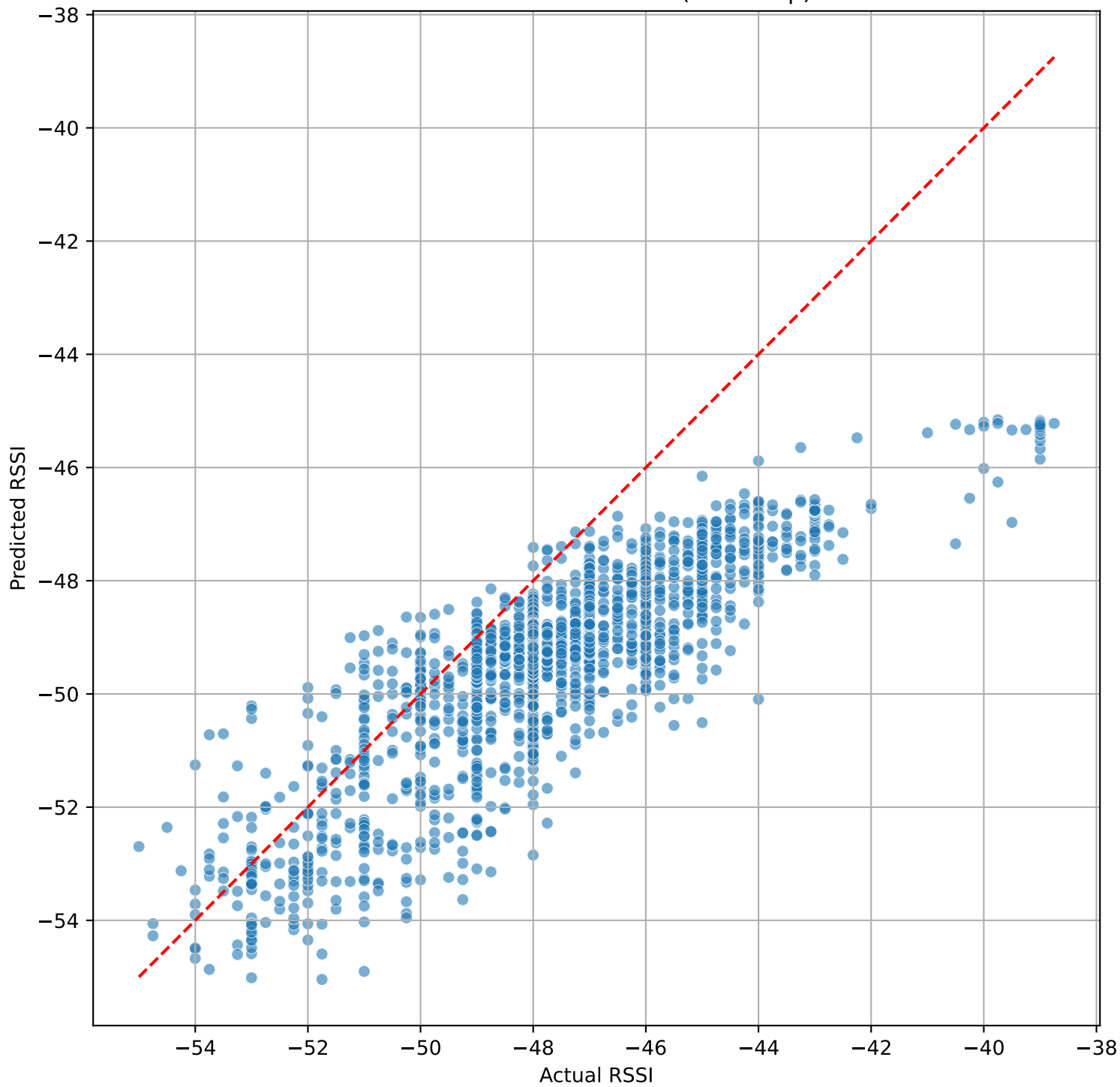
Actual vs Predicted RSSI (First Step)



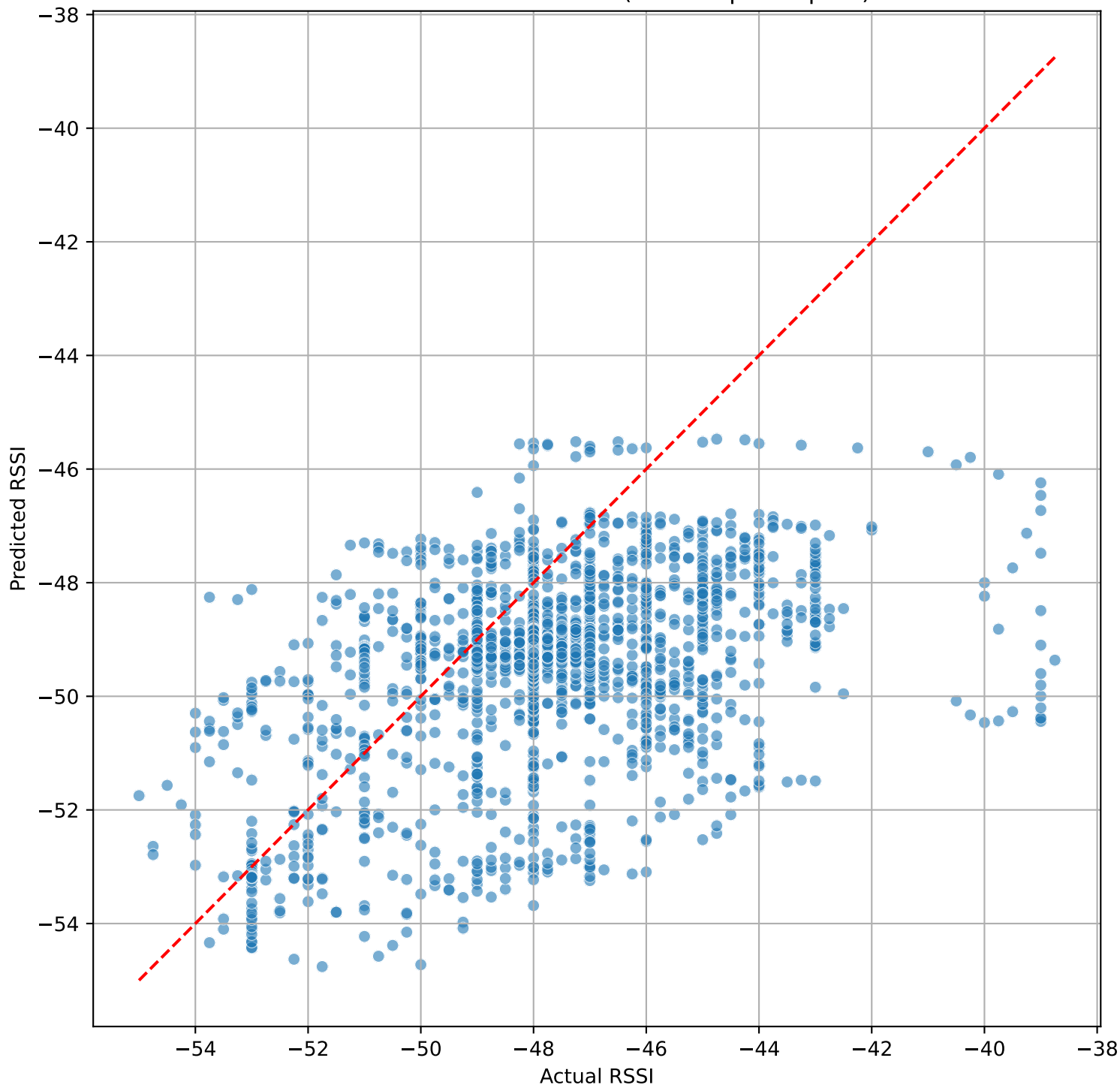
Actual vs Predicted RSSI (Multi-Step (step 20))



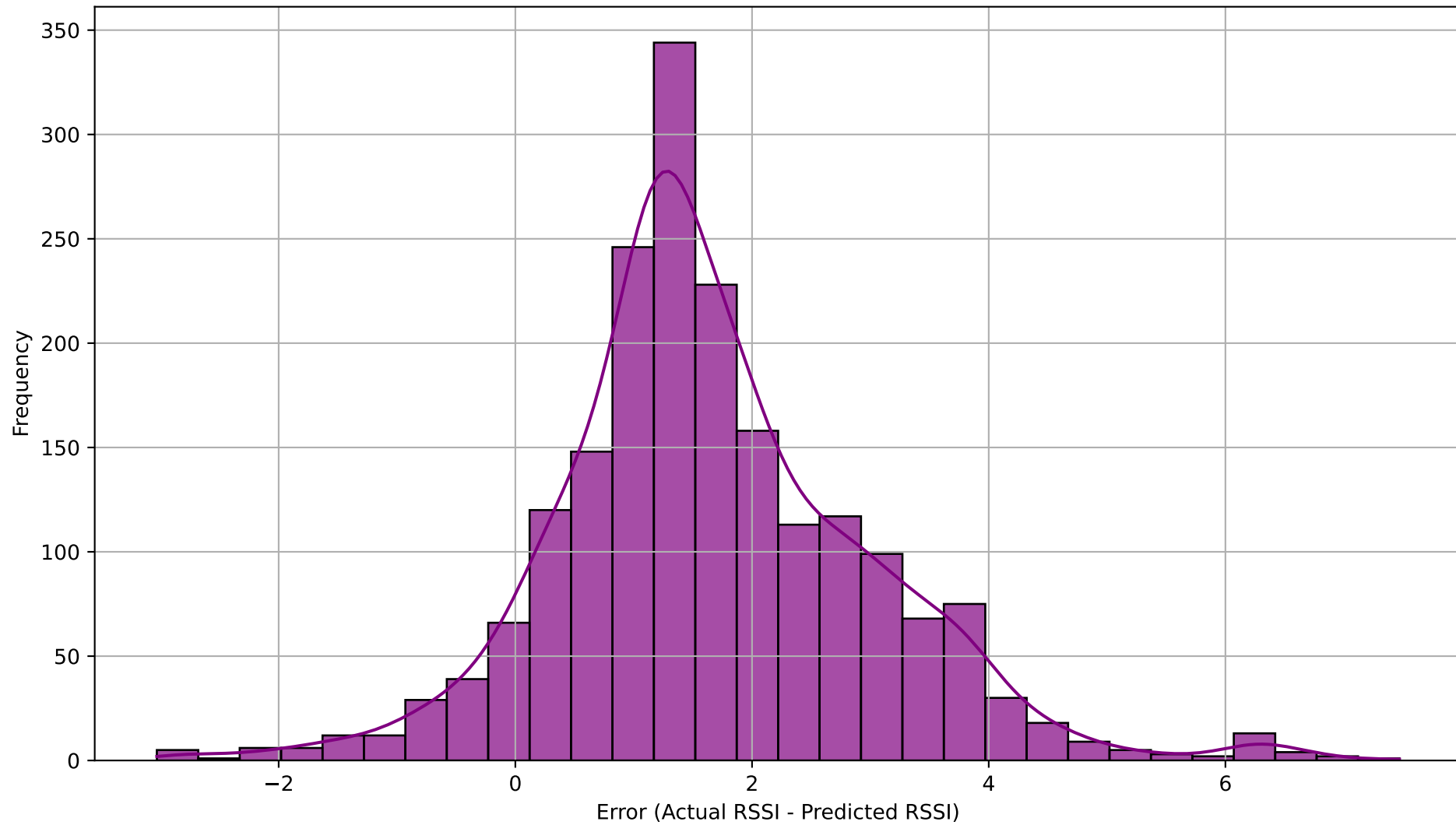
Actual vs Predicted RSSI (First Step)



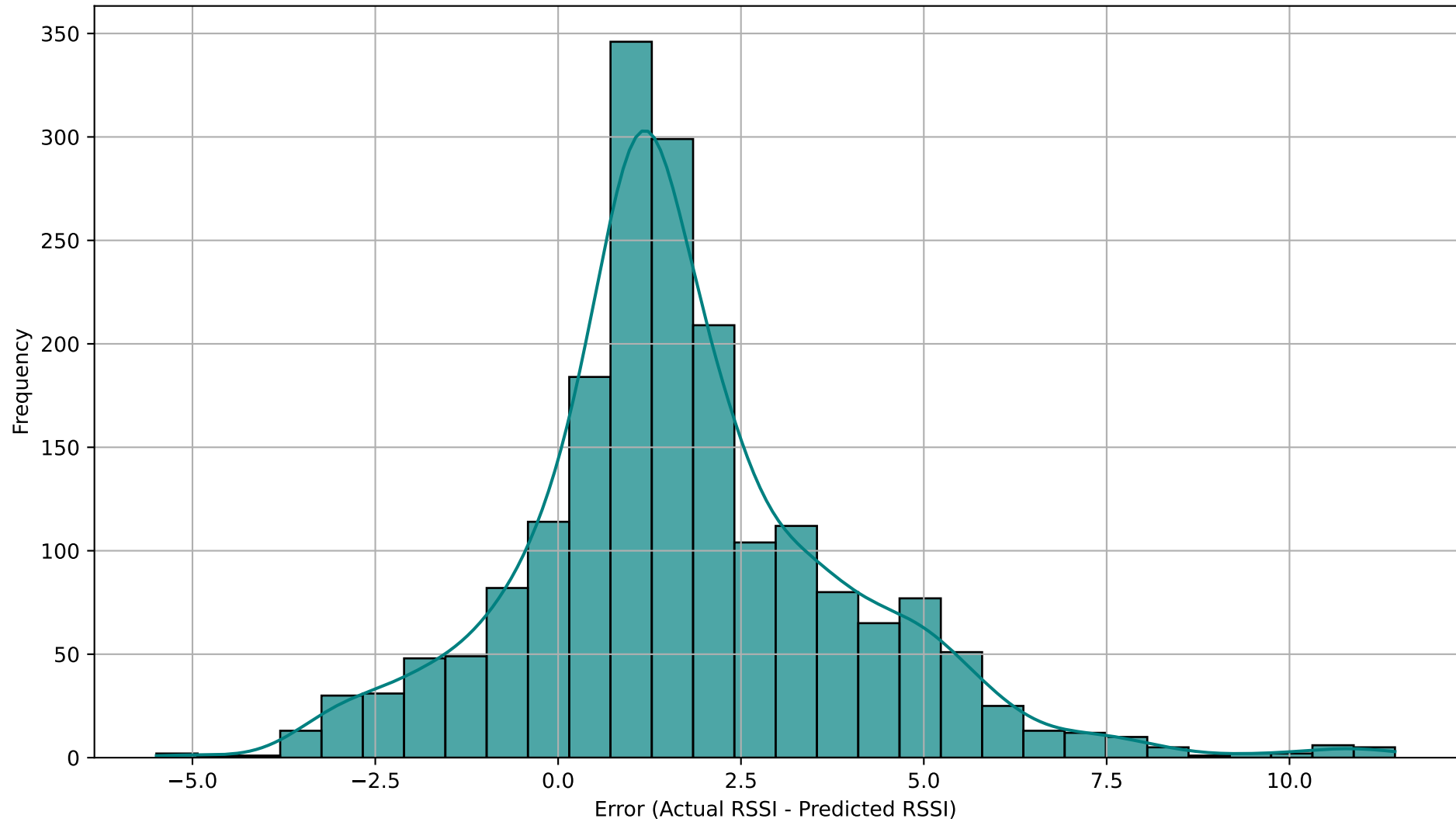
Actual vs Predicted RSSI (Last Step - step 20)



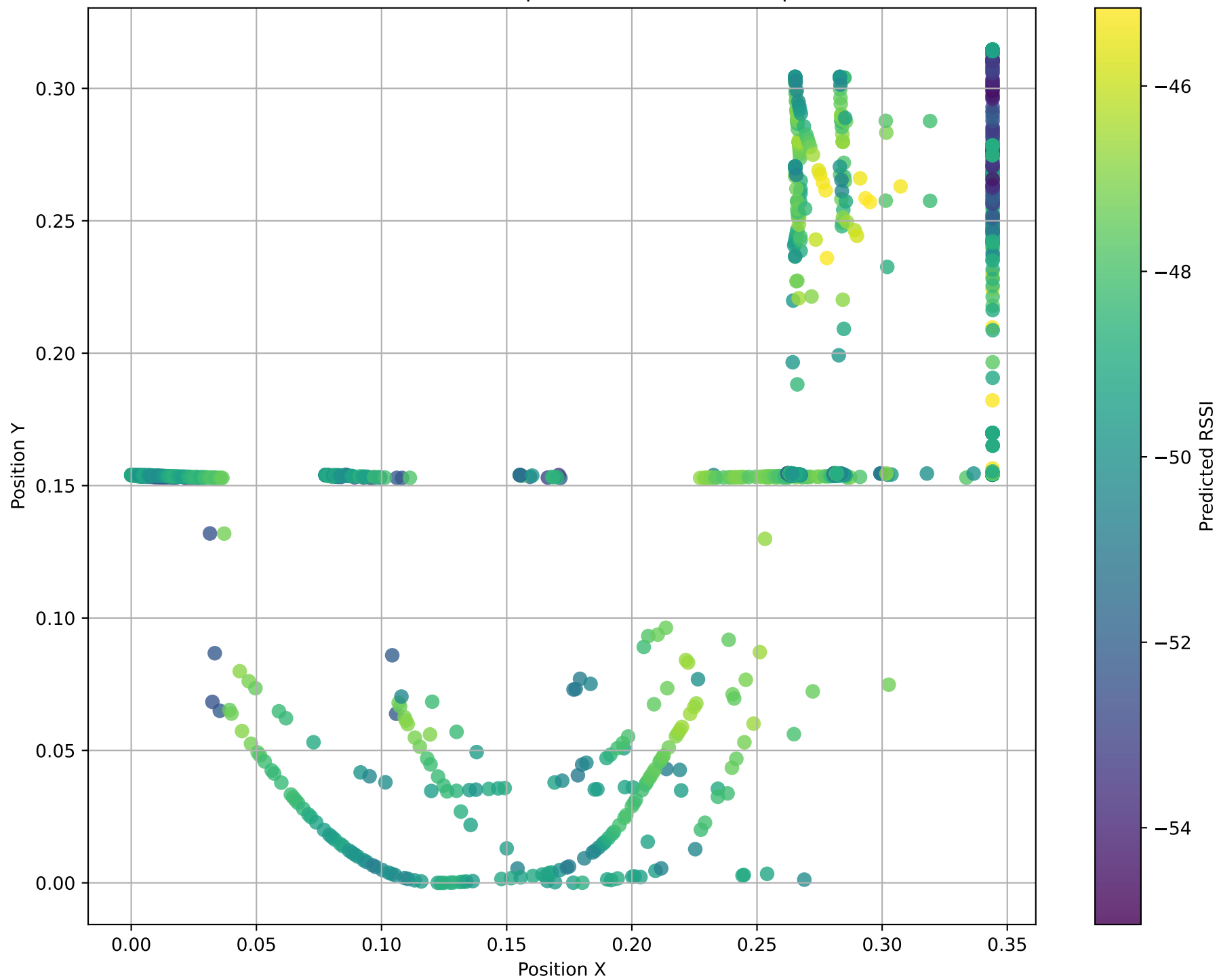
Distribution of Prediction Errors (First Step)



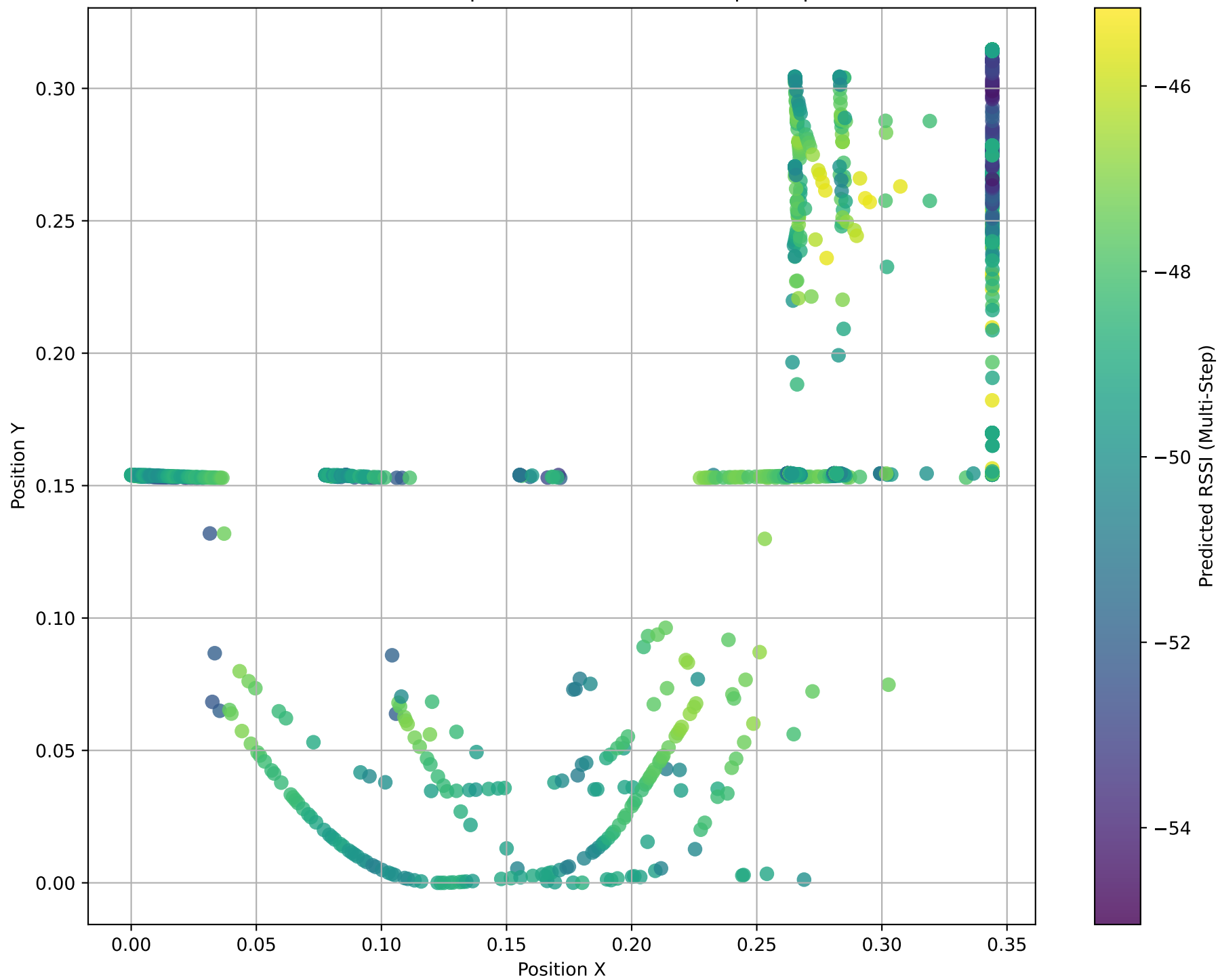
Distribution of Prediction Errors (Last Step - step 20)



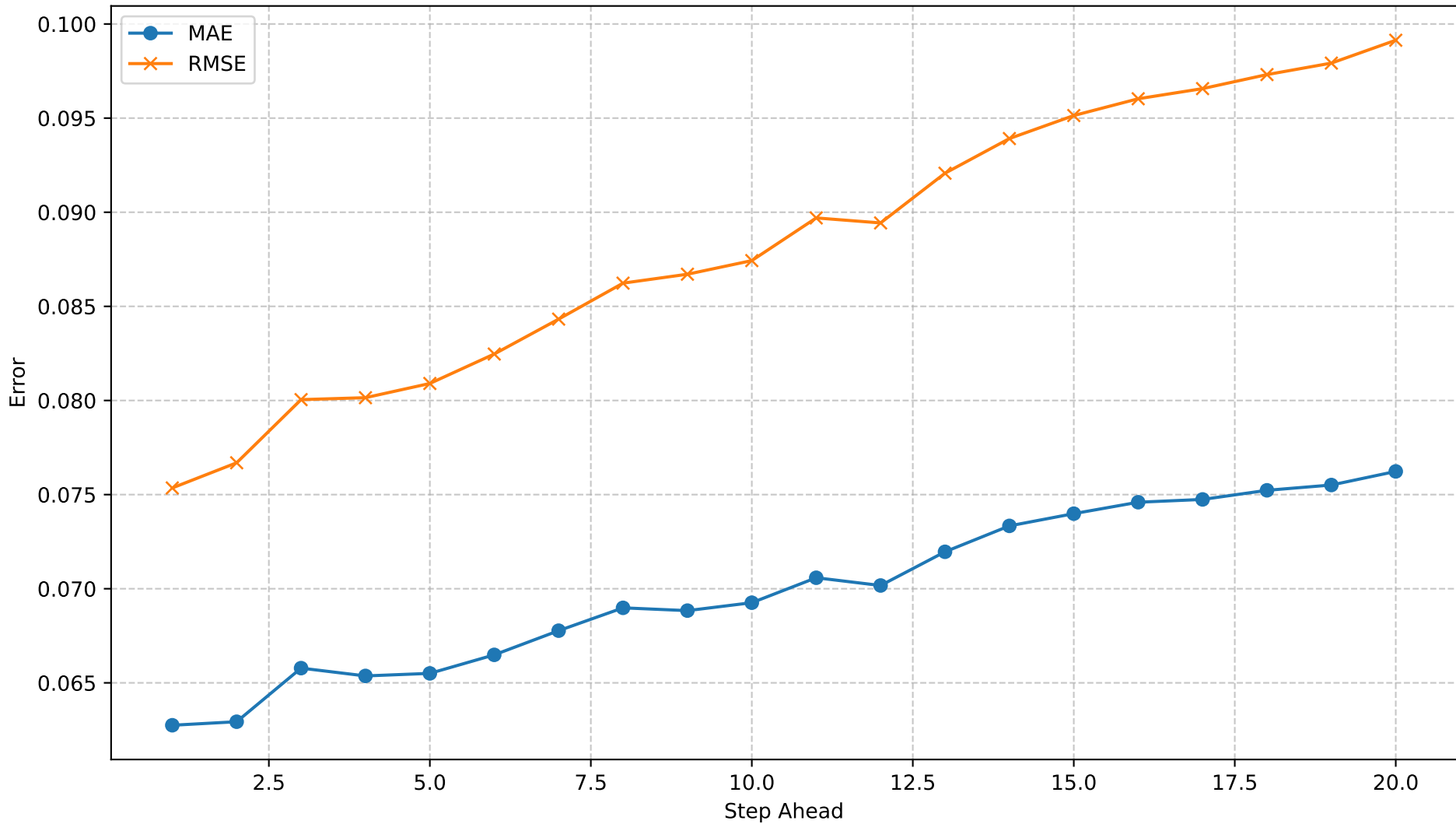
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.06 / 0.08
- Mediana : 0.07 / 0.09
- 20º passo : 0.08 / 0.10

Crescimento do erro: 21.5% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 20 (0.08), 19 (0.08), 18 (0.08)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.